ML-DLOps Assignment 1

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Google Colab Links:
Question 1 -
https://colab.research.google.com/drive/15jMj5_K2rrH81WaOkINfthO4JYMyoqOM?usp=sharing
Question 2 -
https://colab.research.google.com/drive/1Ov4oErNKU0z2_K5bxhefOYxpy49O3IIr?usp=sharing

Github link:
Repository -
https://github.com/arpitadeshmukh/MLOps-Arpita_Abhijit_Deshmukh-B23CM1007/tree/Assignment1
Page -
https://arpitadeshmukh.github.io/MLOps-Arpita_Abhijit_Deshmukh-B23CM1007/

Question 1(a) - CNN Experiments

Trained various models on two different datasets MNIST and FashionMNIST. The following hyperparameters were taken into consideration:

Parameter	Values
Batch size	16, 32
Optimizer	SGD, Adam
Learning Rate	0.001, 0.0001
Models	ResNet-18, ResNet-50
Pin Memory	True, False
Epochs	5, 10

Results:

MNIST Dataset						
Epochs	**PIN MEMORY**	**Batch Size**	**Optimizer**	**Learning rate**	**ResNet-18**	**ResNet-50**
5	True	16	SGD	0.001	**99.34%**	99.13%
				0.0001	98.27%	97.63%
			Adam	0.001	98.92%	98.93%
				0.0001	99.11%	99.05%
5	False	32	SGD	0.001	99.03%	98.93%
				0.0001	96.70%	94.67%
			Adam	0.001	99.18%	98.51%
				0.0001	99.09%	**99.15%**

FashionMNIST Dataset						
Epochs	**PIN MEMORY**	**Batch Size**	**Optimizer**	**Learning rate**	**ResNet-18**	**ResNet-50**
10	True	16	SGD	0.001	91.50%	89.89%
				0.0001	89.71%	84.32%
			Adam	0.001	91.89%	89.84%
				0.0001	91.62%	91.52%
5	False	32	SGD	0.001	90.39%	88.18%
				0.0001	84.40%	79.97%
			Adam	0.001	**92.22%**	**92.03%**
				0.0001	90.35%	89.98%

Observations

- **Dataset:** MNIST is easy, giving very high accuracy with little sensitivity to hyperparameters whereas FashionMNIST is more complex and strongly affected by model and hyperparameter choices.
- **Model:** ResNet-18 matches or outperforms ResNet-50 while training faster; deeper models do not help much for small, low-resolution datasets.
- **Optimizer:** Adam consistently outperforms SGD. SGD degrades at very low learning rates, while Adam remains stable with limited training time.
- **Learning Rate:** LR = 0.001 performs better. LR = 0.0001 leads to slow learning and underfitting within a few epochs.
- **Batch Size:** Smaller batch sizes with Adam give better generalization.
- **Pin Memory:** No effect on accuracy it only improves data transfer speed.
- **Epochs:** MNIST converges quickly whereas FashionMNIST benefits from more epochs.

Question 1(b) - SVM Experiments

SVM was trained on both MNIST and FashionMNIST with varying parameters:

Parameters	Values
Kernel	rbf, polynomial
Degree	2, 3
C	1, 10

Results:

MNIST Dataset				
Kernel	**C**	**Degree**	**Accuracy**	**Time (ms)**
rbf	1.0	-	94.45%	10285.51
rbf	10.0	-	**95.55%**	8977.22
poly	1.0	2	94.15%	8220.28
poly	1.0	3	93.40%	9914.45

poly	10.0	2	95.20%	6378.44
poly	10.0	3	94.10%	8375.44

FashionMNIST Dataset				
Kernel	**C**	**Degree**	**Accuracy**	**Time (ms)**
rbf	1.0	-	86.00%	9880.03
rbf	10.0	-	**87.35%**	10376.90
poly	1.0	2	84.75%	8799.31
poly	1.0	3	82.80%	19842.46
poly	10.0	2	86.80%	8276.69
poly	10.0	3	85.75%	8479.50

Observation:

- **MNIST**: SVM achieves high accuracy (~93–95.5%) across kernels. Increasing C from 1 to 10 consistently improves accuracy.
- **FashionMNIST**: Accuracies are lower (~82–87%) and more sensitive to parameters. RBF kernel with C=10 performs best, while higher-degree polynomial kernels degrade accuracy.
- Polynomial kernels with higher degree increase computation time without improving accuracy.
- SVM performs well on MNIST but struggles more on FashionMNIST due to higher data complexity.

Question 2 - Device Experiments

Trained models on FashionMNIST dataset with varying the following:

Parameters	Value
Device	CPU, GPU
Optimizer	SGD, Adam
Model	ResNet-18, ResNet-50

Other parameters like batch size were fixed to 16 and learning rate fixed to 0.001.

Results:

Device	Optim	Accuracy		Time (ms)		FLOPs	
		ResNet 18	ResNet 50	ResNet 18	ResNet 50	ResNet 18	ResNet 50
CPU	SGD	82.10%	81.01%	8593745.54	26735498.56	1.82e+9	4.13e+9
CPU	Adam	87.20%	88.10%	9342563.67	27645175.89	1.82e+9	4.13e+9
GPU	SGD	84.58%	79.55%	332970.71	1005069.34	1.82e+9	4.13e+9
GPU	Adam	**90.50%**	**90.60%**	346891.90	1038062.73	1.82e+9	4.13e+9

Observations:

- **Device:** GPU training is much faster than CPU, while FLOPs remain the same since they depend on model architecture, not hardware.
- **Optimizer:** Adam gives higher accuracy than SGD across both models and devices due to faster and more stable convergence.
- **Model:** ResNet-50 has much higher FLOPs and training time but does not give consistent accuracy gains over ResNet-18, indicating over-parameterization.